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Assessment of genetic divergence among different species of *Aloe* collected from Jharkhand Ranchi

Kamini Kumar[a]*
*[a]University Department of Botany, Ranchi University, Ranchi, Jharkhand – 834001

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Abstract : Assessment of Genetic divergence using Mahalanobis's D^2 statistics was carried out with the help of karyotype analysis among six species of *Aloe* growing in Ranchi, Jharkhand. The six species of *Aloe* could be grouped into three constellations. The clustering pattern indicated that clustering was due to genetic distance. These distances suggested that the genetic recombination among genotype of these clusters would result in considerable heterosis.

Key Words: Genetic divergence, *Aloe* karyotype, Inter and Intra constellation.

INTRODUCTION

The divergence analysis had performed to find out the diverse genotypes for closeness among the species. Clustering by D^2 statistics is useful for hybridization purposes. However, D^2 statistics (Mahanalobis's generalized distance) has been successfully utilized to bring out the karyotypic differences between different varieties of the same species or different species of the same genus, especially when the differences are limited.

MATERIALS AND METHODS

The plants of six species of *Aloe* were collected from local nurseries and from forest areas of Ranchi, Jharkhand and were taken for the study of genetic divergence. The root apices about 1-5 mm in length were from the pot between 1.00 pm to 2.00 pm and were treated with 0.002M- 8-Hydroxyquinolene at 15ÚC for five hours. They were fixed in fixative (1:3 Acetoalcohol) and then preserved in 70% alcohol. Ten well separated chromosome plates were taken for experimental purpose. The genetic divergence was assessed using D^2 statistics.[1,2] The results were tabulated by calculating:

(a) Absolute length of chromosome.

(b) Relative length of each chromosome.

(c) The angular value of the chromosome.

(d) The log value: the ratio of long arm and short arm was converted into logarithmic scale.

*Corresponding author :
Phone: 09430730756
E-mail : kaminikumar@yahoo.co.in

RESULTS AND DISCUSSION

Karyotype analysis in the present investigation showed that constancy in chromosome number in *Aloe* have not been reported. The different species of *Aloe* under consideration showed asymmetry in their basic chromosome numbers. The three species *Aloe abyssinica* Lam, *Aloe barbadensis* Mill. and *Aloe plicatalis* showed the diploid chromosome number, 2n = 2x =14, whereas, *Aloe harlana* and *Aloe distans* Haw. showed 2n = 3x = 21. A regular tetraploid (2n = 4x = 28) genetic constitution was observed in *Aloe variegate* Linn. A typical bimodal karyotype was found with eight long and six short chromosomes.

The analysis of genetic divergence in the different species of *Aloe* (table 1&2) under investigation indicated that all the species could be grouped into three constellations (fig.1). The constellation CI consisted of three species $V_{2,}$ V_4 and $V_{6,}$ which represented *A. barbadensis* Mill., *A.harlana* and *A.variegata* Linn. The constellation CII comprised of two species *Aloe abyssinica* Lam, and *A. distans* haw.and CIII consisted only one species V_3 i.e. *A. plicatalis*. The analysis made in view to

genetic divergence indicated that intra constellar distance in the constellation CI, CII and CIII were 3.738, 1.820 and 0.00. The inter constellar distance between CI – CII, CII – CIII and CI – CIII were 1.994, 2.551 and 1.411 respectively. The highest and lowest D^2 values in their interconstellar divergence was observed between CII – CIII and CI – CIII respectively (Fig.-1).

The divergence analysis [1-3] had performed to find out the diverse genotypes for closeness among the species. Clustering by D^2 statistics is useful in hybridization purposes. However, D^2 Statistics has been successfully utilized to bring out the karyotypic differences between different varieties of the same species or different species of the same genus, especially when the differences are limited. The evidence arrived at using this type of analysis has supported the conclusion based on morphological characters. [4-11] The grouping in the present study is based on the degree of divergence obtained by pooling the square of distance for 14 (or 21 or 28) chromosomes between a pair of species tentatively, so that intra – group D^2 values

Table-1 : K – Value of individual Chromosome of six species of *Aloe*.

Species under consideration	Chrom I	Chrom II	Chrom III	Chrom IV	Chrom V	Chrom VI	Chrom VII
Aloe abyssinica Lam V_1	1.86	1.93	2.39	2.88	3.18	2.71	3.32
A. barbadensis Mill. V_2	1.78	1.87	1.96	1.97	3.52	3.16	2.84
A. plicatalis V_3	1.59	1.38	2.05	1.31	3.55	3.32	2.27
A. harlana V_4	1.63	1.70	1.65	1.77	2.40	3.04	3.04
A. distance Haw. V_5	2.23	2.40	2.74	3.32	2.23	2.31	2.08
A. variegata Linn V_6	1.77	2.02	1.98	1.69	2.40	3.31	3.43

Table-2 : Generalised distance and clustering pattern of six species of *Aloe*.

S. N.	V_1	V_2	V_3	V_4	V_5	V_6
1.	V_2-1.365	V_3-1.094	V_2-1.094	V_6-0.437	V_1-3.313	V_4-0.437
2.	V_6-1.922	V_1- 1.365	V_4-2.429	V_2-1.589	V_4-5.798	V_2-1.712
3.	V_4-2.399	V_4-1.589	V_6-3.291	V_1-2.399	V_2-5.910	V_1-1.922
4.	V_5-3.313	V_6-1.712	V_1-4.252	V_3-2.429	V_6-6.455	V_3-3.291
5.	V_3-4.252	V_5-5.910	V_5-8.764	V_5-5.798	V_3-8.764	V_5-6.455

* (In the table serial number 1. Is closet and chronologically serial number 5 is farthest)

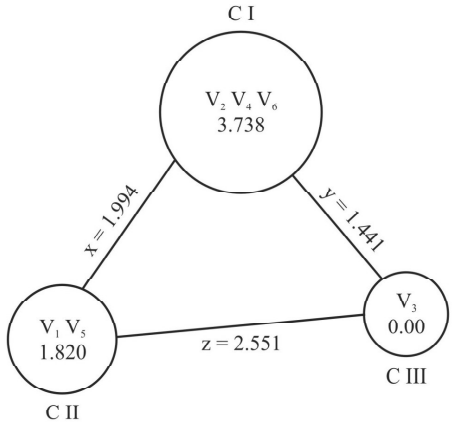


Fig.-1 : Generalised distance between and within constellations on the basis of D^2 - values among six species in *Aloe*.

are much smaller than inter group values. Though this grouping is empirical it gives fairly good idea of the relationships or differences (Fig.-1).

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